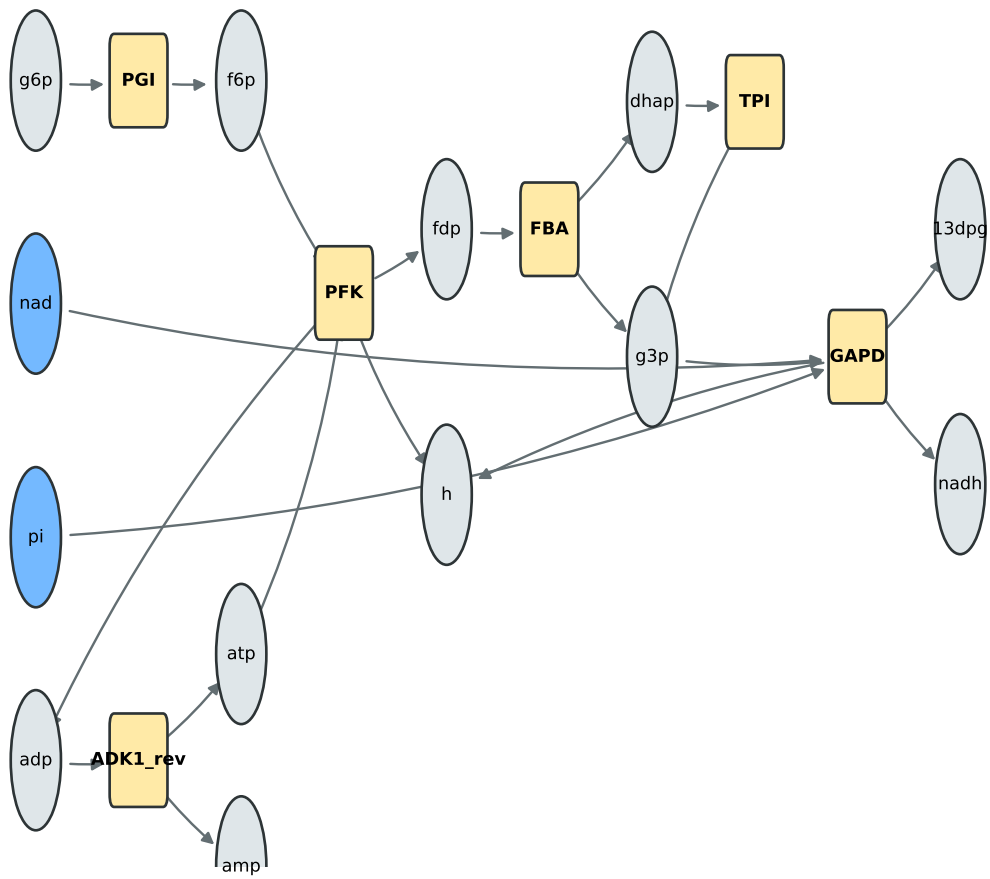
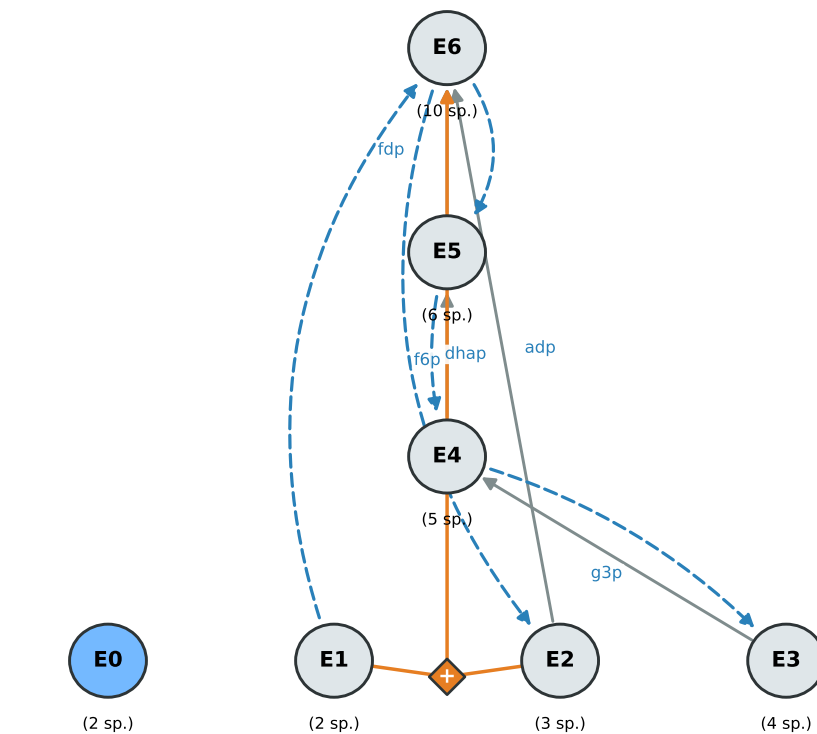


(a) Illustrative reaction network
(6 real *E. coli* reactions: PGI, ADK1, PFK, FBA, TPI, GAPD)



● food species ● species ■ reaction

(b) Fundamental ERC hierarchy
(containment, 1 synergy, complementarity chain)



E0 = food closure
(nadh, pi) — a separate
persistent ERC

— containment
◆ fundamental synergy
- - - fundamental complementarity